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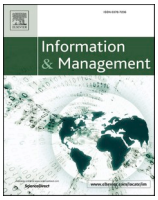

Two faces of freemium strategy in social games: The interplay between perceived enjoyment, envy, and in-app purchase intention

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ABSTRACT

In freemium social games, disparities between paid and free players could evoke negative experiences, particularly when paid players gain significant advantages by purchasing in-game items. Drawing on affordance theory and social comparison theory, this experimental study reveals that both positive (e.g., perceived enjoyment) and negative (e.g., benign envy) experiences strengthen players' willingness to make in-app purchases. Perceived enjoyment is positively associated with skill–challenge balance, social interaction, and control, while both benign and malicious envy are induced by fairness. Notably, a state of skill–challenge imbalance is also found to drive perceived enjoyment, providing a critical extension to the existing literature on enjoyment experiences in gaming contexts.

1. Introduction

Over the past decade, online social games have experienced substantial growth. Unlike single-player games, social games enable users to communicate and interact directly with other players, offering an experience that closely mirrors real-world social interactions. Notable examples include massively multiplayer online role-playing games (MMORPGs), such as World of Warcraft, and social network games such as FarmVille.

Freemium games, a type of online social games, have gained significant popularity in recent years. In contrast to traditional PC or console games, which require an upfront purchase, freemium games operate on a freemium business model. This model allows users to download and play the games for free [1–3], with payment only required for acquiring additional virtual goods or premium services, such as extra lives, character clothing, or high-powered weapons and tools [4–6].

Despite ample growth opportunities, a critical challenge for the freemium game model lies in its high operating costs, largely driven by the disproportionate number of free users compared with the relatively small proportion of paying users [7,8]. Research by Paavilainen et al. [9] indicates that only 1 % to 5 % of users playing freemium social games opt to purchase in-app items. Similarly, in sectors beyond gaming, such as business-to-business (B2B) and business-to-consumer (B2C) software companies employing freemium models, fewer than 5 % of users typically transition from free tiers to premium, full-priced products [10]. This conversion rate is even lower in the emerging mobile gaming market, where only 2 % of players are willing to pay, as highlighted in a recent report by Liftoff [11]. Rather than relying on small payments from all players, the freemium game model generates the majority of its revenue from a small subset of users. Players in this freemium model are typically categorized as whales (constituting less than 10 % of users but contributing over 50 % of revenues), dolphins, and minnows, the latter composing more than 50 % of the user base but providing minimal or no financial contribution [12]. Despite a few fabled cases in this sector, many freemium games fail to sustain their market presence, often disappearing quickly. This is largely due to the extremely low switcAhing costs and the lack of player commitment, which make it easy for users to migrate to newer games. Understanding how to enhance users' willingness to make in-app purchases in the freemium game context effectively is therefore of critical importance.

Sweetser and Wyeth [13] identify player enjoyment as the primary goal of computer games. Perceived enjoyment is widely recognized as a key factor in shaping users' gaming intentions and behavior [14,15]. However, its relationship to players' in-app purchase intentions in the context of freemium games remains insufficiently understood [3,16,17]. This study seeks to address this gap by examining how perceived

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enjoyment, as a positive emotional experience, influences in-app purchase intentions within freemium social games.

The distinctive feature of social games is the presence of other players, which enables hierarchical relationship comparisons [18,19]. These hierarchical relationships distinguish leaders from followers based on gaming abilities or specific ranking criteria. Within such an environment, players naturally engage in social comparisons by evaluating their performance relative to that of others [18,19]. Recent research by Esteves et al. [15] has investigated the influence of social comparisons on players' perceived enjoyment, online gaming self-efficacy, and game continuance. Their findings confirm that elements of social comparison significantly impact players' engagement and continued participation in social games. In the case of freemium social games, free players form negative perceptions of those who purchase in-app items with real money, labeling them as "cheaters," particularly when the advantages gained by paid players are visible [5]. This circumstance has sparked widespread debates regarding the fairness of virtual gaming goods. Feelings of unfairness are triggered when players perceive that those who spend real money on in-game items gain significant advantages, making them hard to defeat [9,20]. As one interviewee in *Wired* magazine noted, many top leaderboard positions are achieved through the purchase of additional items rather than through skill alone [21]. This gap between free and paid players induces envy, a form of negative emotion [22]. Smith [23] suggests that envy tends to occur when one perceives that the envied does not deserve the advantage. In this sense, we expect that fairness could serve as a bridge to explain and understand individuals' feelings of envy. However, the specific impact of fairness on players' envy has yet to be validated empirically within the context of freemium social games.

Furthermore, the extant literature also shows that envy is one of the important driving forces behind individual consumption [24,25]. This concept has been extensively leveraged within the video game industry to promote sales. For instance, in SimCity Social, players observing others' more developed cities are often motivated to spend real money on advanced features, enabling them to upgrade their cities more easily [26]. Hence, there is a need to investigate the underlying mechanisms of players' envy and its relationship with in-app purchase intentions in the context of freemium social games.

Based on the above discussion, freemium social games can trigger both positive and negative emotional responses. While enjoyment remains the most frequently reported positive emotion associated with gameplay, negative experiences have received comparatively little scholarly attention, despite their prevalence. As highlighted, the distinctive features of freemium social games often facilitate social comparisons among players, potentially leading to negative emotions such as envy. In this study, we propose examining perceived enjoyment as a representation of positive experiences and envy as a manifestation of negative emotions arising during gameplay. Although a growing body of literature has explored various aspects of freemium social games [17,18,20,27–29], the simultaneous investigation of both positive and negative experiences, particularly in the context of their influence on in-app purchase intentions, remains underexplored. This study seeks to address this gap by providing a comprehensive analysis of this dual emotional dynamics in freemium social games.

We grounded our study in affordance theory and social comparison theory to investigate two primary factors influencing in-app purchase intention – perceived enjoyment and envy. Affordance theory refers to the potential actions or possibilities that a technological object provides to users [30]. It is frequently employed to explore the relationship between technology and user experiences, emphasizing how the characteristics of technology shape user interactions and the value derived from systems or devices [31,32]. The interplay between game affordances, player experiences, and behavioral intentions has been studied widely within the context of online gaming [28,33–35]. In the case of freemium social games, we identified four key affordances: skill–challenge balance, control, social interaction, and fairness. These affordances were analyzed to understand their roles in shaping both positive and negative emotional experiences during gameplay and, ultimately, their influence on players' in-app purchase intentions. The second theory is social comparison theory, which is drawn upon to explain the negative emotions players may experience as a result of in-app purchases. Freemium social games are expected to amplify players' tendencies to compare their performance with that of others, often resulting in feelings of envy, particularly toward those who attain higher ranks primarily through purchasing in-app items. By incorporating affordance theory and social comparison theory, this study aims to uncover the underlying mechanisms linking game affordances to players' emotional responses, which, in turn, influence their in-app purchase intentions. This integrative approach offers a more comprehensive understanding of in-app purchase behavior in the context of freemium social games.

2. Theoretical background

2.1. Affordance theory and game affordances

Affordances refer to the opportunities or possibilities that an object offers or provides to people [36]. This concept has been applied widely in research on technology and user interaction. Suh et al. [37] emphasize that the core principle of affordance theory lies in its relational nature, wherein technological capabilities are not limited to predefined functions or properties but instead arise from the interaction between a user and technological artifacts within a specific context. In other words, affordance depends on the interplay between a game's characteristics and a player's ability to take action. Affordance theory thus serves as an analytical framework for connecting the technological features of mobile games with players' experiences [38,39].

The relationship between game affordances and behavioral intention has been explored widely in existing literature. Building on this foundation, we propose that affordance theory can be applied to explain the connection between players' experiences and their purchase intentions in the context of freemium social games. In massively multiplayer online games (MMOGs), affordances bridge users' potential actions with technical elements, allowing them to utilize various features to support their goal-oriented actions [30]. Previous studies on MMOGs have focused primarily on affordances such as achievement, social interaction, and immersion to examine game addiction through the lens of hedonic management processes [40]. Similarly, Wang et al. [27] have investigated the indirect effects of game fairness and the balance between challenges and skills on players' in-game purchase behavior. Drawing on the affordance theory and social capital literature, Gong et al. [41] explored how social network embeddedness (SNE) technology affordances, specifically identity transparency and information transparency, are integrated into games, influencing players' social experiences and game performance. Wang et al. [28] investigated in-game purchase behavior in multiplayer online battle arenas (MOBA), focusing on game orientation and identification through two types of affordances: instrumental and hedonic. Additionally, indirect links between MMOG affordances—such as achievement, interaction, identity, and immersion affordances—and addictive use have also been indicated via virtual-domain perfectionism in seeking excellence and avoiding failure [33].

This analysis highlights the diverse affordances offered by MMORPGs. A review of the existing literature on the affordances of freemium social games reveals inconsistencies in terminology, with identical or similar affordances labeled differently, and multiple technical features can be grouped under a single affordance. To address this, we categorized game design elements with shared characteristics into broader affordance categories. Previous research on MMOGs has focused predominantly on two primary types of game affordances: instrumental and hedonic [28]. These affordances are particularly pertinent to player orientations in free-to-play MOBA games, aligning

with the distinct motivations of "play-to-win" and "play-for-fun" players. Instrumental affordances are linked to game design elements that primarily support players' instrumental goals, such as achieving success or "play-to-win" objectives. In contrast, hedonic affordances cater to players' hedonic goals, such as enjoyment and "play-for-fun" experiences. Specifically, instrumental affordances include achievement affordance, game fairness, and the balance of challenges and skills [27,28,33,40], while hedonic affordances encompass social interaction, immersion, identity affordance, game novelty, and design aesthetics [28,33,40]. Building on the aforementioned summary of game affordances typically explored in MMORPGs, this study identifies four key affordances that shape player experiences in freemium social games: skill–challenge balance, control, social interaction, and fairness.

In MMORPGs, players compete directly against rivals, with each game culminating in a definitive outcome, either a win or a loss. Due to the competitive nature of MMORPGs, players typically achieve success by overcoming various challenges. Sweetser and Wyeth's GameFlow model [13] identifies eight dimensions of player enjoyment, highlighting challenge as a critical component of effective game design. To motivate players, goals must be both attainable and challenging, requiring maximum effort for their achievement [42]. An attainable goal should strike a balance between a high probability of success and the necessity for substantial effort [43]. Regarding the challenge aspect, a well-designed game should be "sufficiently challenging, match the players' skill level, vary in difficulty, and maintain an appropriate pace" [13]. Based on these considerations, we propose the balance of challenges and skills as a fundamental game affordance that supports players' instrumental orientations.

Challenge is defined as a game attribute derived from various design elements, including game design principles and heuristics (e.g., clear goals), design patterns and mechanics (e.g., time constraints), and interface design patterns (e.g., badges and leaderboards). Skills, on the other hand, refer to a player's ability to execute actions within the game [27]. Skill–challenge balance refers to the alignment between the challenges presented in a game and the players' abilities [28]. Building on this concept, we define skill–challenge balance affordance as the game's capacity to maintain an equilibrium between its level of difficulty and the players' skillsets. Optimal player engagement occurs when skills and challenges are balanced; an imbalance—where skills exceed challenges or challenges exceed skills—can lead to boredom or anxiety, respectively [44]. Choi and Kim [45] further highlight that players are more attracted to games that align with their skill levels and provide ample opportunities to utilize those skills effectively.

Control affordance refers to the integration of features that enable players to engage with and manipulate game objects and properties using available physical control mechanisms [46,47]. These control features encompass various input methods, such as keyboard/mouse/joystick combinations, touch screens, and motion controls, as well as functionalities related to game progress saving and the management of in-game resources [48]. These elements are tied closely to a player's perceived ability to succeed in the game. Specifically, they facilitate key skills such as self-control, pattern recognition, decision-making, critical thinking, and memory recall, all of which are essential for managing and understanding game elements effectively [47]. By incorporating features that support control and manipulation, games enhance players' chances of earning rewards and achieving objectives. Thus, we identified control affordance as a critical aspect of game design that aligns with and supports instrumental gamer orientations.

In social games, players' social connections are pivotal. Unlike other online or casual games, which are often solitary experiences, freemium social games incorporate features that actively promote collaboration or competition among players. This distinction creates a unique dynamic of player interaction, setting social games apart from other types of online games. Given the intrinsic emphasis on player interactions in freemium social games, we identified social interaction as a key game affordance aimed at fulfilling hedonic gamer orientations [41]. This concept is also referred to as "social affordance" in the study by Lee et al [40], which describes the extent to which MMOGs facilitate player engagement with others within the virtual world. In freemium social games, various functions, such as chatting, voice communication, and the sharing or exchanging of game elements, are integrated to promote social interactions [49]. These affordances serve as the foundational rules and mechanics that enable players to participate in both competitive and cooperative activities within virtual gaming environments [50,51].

Compared with other types of games, such as role-playing games and traditional social games, the defining feature of freemium social games is the option for players to purchase in-app items. The competitive nature of these games drives game operators to ensure fair competition between free and paying players. Consequently, players in freemium social games place considerable importance on game fairness, expecting equitable matchups with their opponents. Building on this, we propose that game fairness serves as a critical game affordance that supports players in achieving their instrumental goals in freemium social games. According to Rong et al [52], game fairness is conceptualized as a game vendor's commitment to fostering a sense of "justice," enabling players to perceive the gaming environment as equitable and conducive to achieving their objectives. In freemium social games, fairness is particularly salient for free players, as it manifests in various aspects, including resource allocation, game rules, and tactical strategies. Players are less likely to engage with games where others appear to hold disproportionate advantages, such as access to powerful weapons acquired through monetary investment [20]. For example, wealthier players may purchase superior in-game items, granting them significant competitive advantages over others. This raises critical concerns about whether in-game performance and achievements genuinely reflect player skill or are predominantly influenced by financial expenditure. Furthermore, research by Lin and Sun [53] highlights that players are more inclined to abandon games perceived as overly monetized. These findings emphasize the imperative for game operators to uphold fairness affordances, ensuring that all players feel equally valued and treated [27,28,54].

2.2. Social comparison theory and envy

Social comparison theory is a well-established framework for examining comparative behavior in both physical and digital contexts [55]. Social comparison refers to the process through which individuals evaluate and contrast specific aspects of themselves—such as abilities, physical attributes, or accomplishments—with those of others [56,57]. This process can occur in two primary modes: (1) upward comparison, where an individual compares themself to someone perceived as superior, and (2) downward comparison, which involves assessing oneself against others perceived as inferior [58–60]. Given their opposing directions, upward and downward comparisons elicit distinct emotional responses [61]. Upward comparisons often evoke negative emotions such as resentment, envy, or inadequacy in the person making the comparison. Conversely, downward comparisons can generate positive emotions, including satisfaction and gratification, when individuals recognize their superior status.

Envy can thus be understood as a consequence of upward comparison [62–64]. It is defined as a negative emotional response to another person's superior quality, achievement, or possession, where the individual experiencing envy either desires the advantage for themself or wishes that the envied individual did not possess it [65,66]. Envy can be classified into two types: dispositional and situational envy. Dispositional envy refers to a stable personality trait characterized by a general tendency to feel envy in various contexts, often manifesting as persistent negative emotions related to others' advantages [67]. In contrast, situational envy arises from specific circumstances, where the envier experiences negative feelings triggered by unfavorable comparisons in a particular situation [68]. Building on prior studies on social media usage

[24,62], purchase intention [25], and non-functional item consumption in free-to-play mobile games [69], our research specifically investigates the role of situational envy in shaping user behavior. This study examines situational envy among players within the context of social freemium games through an experimental game setting. The social features of freemium games not only facilitate player interaction but also draw players' attention to their partners' scores and rankings. We hypothesize that these dynamics increase players' propensity to compare their performance with others, ultimately evoking feelings of envy—particularly toward those who achieve higher ranks primarily through the purchase of in-app items.

Existing literature has examined the prerequisites and mechanisms through which individuals experience envy. In Festinger [56]'s seminal work on social comparison theory, it is observed that individuals acquire diagnostic information for self-evaluation by selecting specific standards for comparison, particularly on aspects that are highly relevant to them. Smith [23] identifies four conditions necessary for envy to arise: (1) similarity between the individual and the compared person, (2) recognition of the situation's relevance to oneself, (3) perception that the envied object is difficult to obtain (i.e., low perceived control), and (4) belief that the envied person does not deserve their advantage. According to Festinger's proposition on social comparison [56], first, individuals are more likely to make comparisons with others who share similar attributes or background characteristics, such as gender, age, or social class [56,70,71]. For example, Hill et al. [72] found that individuals are more likely to envy someone of the same sex rather than the opposite sex. Similarly, Liberman et al [73] observed that close friends and acquaintances with a low degree of social distance are common targets for comparison. In the context of social games, shared characteristics—such as being partners in the same game—can make individuals more prone to envy. Envy is more easily triggered by those within the same social or gaming environment. Second, when a person perceives that unfavorable conditions are particularly relevant to their own situation, the likelihood of envy increases. For example, players who have lower performance than their counterparts may notice that in-app purchases play a disproportionately significant role in determining final scores and rankings. Consequently, envy is likely to be elicited when players recognize that higher ranks are primarily achieved through the purchase of in-app items using real money. Third, perceived control refers to an individual's perception of their ability to manage a situation. Envy is more likely to arise when a person perceives that unfavorable conditions are difficult to overcome [74] and that desired attributes are unattainable. In the context of our study, the envy experienced by players who do not possess in-app items stems from the belief that it is nearly impossible to outperform other players or achieve top ranks without purchasing in-app items. Last, perceived deservingness involves an individual's assessment of the alignment between a situation and its outcome. When a perceived match exists, the advantages others possess are considered justified; when there is no match, they are seen as undeserved [75]. This evaluative process, which entails reassessing others' advantages and one's own disadvantages, plays a crucial role in determining the type of envy experienced. Benign envy arises when the envied individual is perceived to deserve their advantages, such as achieving high ranks due to superior gaming skills. In contrast, malicious envy is triggered when advantages are perceived as unjust, for example, when high performance is attributed solely to the acquisition of in-app items.

Envy can be categorized into benign envy and malicious envy [66,76]. The two types share common characteristics, including intense feelings of inferiority and displeasure directed toward another individual. However, they differ in their attentional focus and motivational consequences. Malicious envy prompts individuals to focus on the envied person, whereas benign envy emphasizes self-improvement and the means of enhancing one's own abilities or status [77]. These differing attentional foci lead to distinct motivational outcomes. Malicious envy fosters a "pulling-down" motivation, where individuals may engage in undesirable behavior toward the envied person, such as degrading them, depriving them of opportunities to achieve their goals, causing harm, or wishing misfortune upon them [66,78,79]. In contrast, benign envy fosters a "moving-up" motivation, encouraging the envier to enhance their performance and achieve what they perceive as lacking compared to the envied individual [80]. Although envy evokes negative emotions, marketers often seek ways to trigger envy among consumers, as it can drive consumption [81]. Observing others with superior possessions can create a strong desire to acquire similar goods [82,83]. For instance, Taylor and Strutton [25] found that Facebook usage stimulates envy among users, which subsequently leads to conspicuous consumption. Therefore, it is reasonable to suggest that the emotion of envy can significantly increase a person's purchase intentions.

It is important to note that envy does not necessarily lead to identical responses across different situations [80]. For instance, in SimCity Social, an online game where players build their own cities, both benign envy and malicious envy may arise from observing others' more developed cities. However, these two types of envy lead to distinct actions. Players experiencing benign envy may be motivated to spend money on exclusive resources to enhance their own city construction, whereas those experiencing malicious envy might engage in destructive behavior aimed at sabotaging others' cities [26]. Therefore, it is crucial to differentiate between these two forms of envy [84]. This study investigates the distinct roles of benign and malicious envy in the context of social games and examines whether fairness in freemium social games differentially influences these types of envy. Furthermore, it explores how these emotions subsequently impact users' in-app purchase intentions.

2.3. Perceived enjoyment

The motivation perspective suggests that the primary drivers encouraging individuals to use information technology are intrinsic and extrinsic motivation [85]. Intrinsic motivation is centered on the enjoyment and satisfaction derived from engaging in an activity, while extrinsic motivation stems from the expectation of achieving external goals [86]. Perceived enjoyment is commonly defined as "the extent to which the activity of using a specific system is perceived to be enjoyable in its own right, independent of any performance outcomes resulting from system use" [85]. Previous studies have established a significant relationship between perceived enjoyment and users' acceptance of hedonic systems [85], indicating that individuals are more likely to adopt a system if they perceive its use as enjoyable.

While traditional task-oriented technologies prioritize the achievement of specific goals or performance improvements, entertainment-oriented technologies focus on providing users with pleasure and enjoyment. Online games are a prime example of the latter, where enjoyment is often regarded as the most critical objective of gameplay, as argued by Shin [87] and Wei and Lu [88]. Existing literature has empirically demonstrated a positive relationship between enjoyment and both behavioral intention to engage with and continued intention to use such technologies [15,89,90]. In the context of massively multiplayer online role-playing games (MMORPGs), studies by Lee et al [40] and Tan and Yang [17] have similarly highlighted the importance of enjoyment as a central motivator. Building on this foundation, we examine perceived enjoyment as a primary driver for players of freemium social games.

3. Hypothesis development

3.1. Skill–challenge balance toward perceived enjoyment

Skill reflects a player's ability to execute actions within a game [91]. Existing literature on mobile gaming provides substantial evidence of the positive relationship between skill levels and enjoyment [92,93]. On the other hand, game challenge refers to the experience of having one's

abilities pushed to their limits [94]. Players derive enjoyment from gaming when they successfully overcome positive challenges [95]. However, since mobile games are often played as a leisure activity, overly difficult games may fail to provide an enjoyable experience and serve their intended purpose as a relaxing pastime [96].

Hoffman and Novak [44] emphasize that a balance between a player's skills and a game's challenges is essential for achieving a state of flow [97]. Conversely, any imbalance can evoke feelings of boredom or anxiety [98]. Recent studies have emphasized the importance of balancing game challenges and skills as a critical determinant of game playability [52,96], which significantly influences perceived enjoyment. Prior research has confirmed that achieving a balance between perceived skills and challenges enhances players' immersion in and enjoyment of a game [27,99–101]. Based on these findings, we propose the following hypothesis:

Hypothesis 1. Skill–challenge balance influences perceived enjoyment positively.

3.2. Control toward perceived enjoyment

Control is defined as an individual's perception of their ability to manage a specific interaction [102]. In the gaming context, control refers to a user's capacity to influence, guide, and dominate in-game events [103]. Players who perceive greater control over the game experience a sense of freedom and autonomy, enabling them to act as they wish within the gaming environment.

When users engage with specific services or products, they are more likely to derive significant hedonic value if they believe they have control over those services or products. Previous studies have explored whether users' perceived control is a key determinant of their perceived enjoyment in gaming contexts. For instance, Limperos et al. [104] demonstrated a positive relationship between perceived control and enjoyment through a between-subjects experimental design using two game consoles. Findings from human–computer interaction literature further support this notion, suggesting that gamers feel more engaged and fascinated when they are given greater control over the gaming experience [105]. Higher levels of perceived control have been shown to enhance users' satisfaction and pleasure [106]. In contrast, low perceived control combined with a high level of difficulty is more likely to evoke feelings of anxiety and dissatisfaction among users [107].

In the context of gaming, control enables players to execute various tasks effectively within the game [108]. In MMORPGs, control is facilitated through features such as shortcut bars, macro commands, and controllers [109]. The availability of such features simplifies character management and task completion, thereby enhancing players' overall gaming experience [13,109]. Existing literature has provided empirical evidence supporting the role of control in driving game enjoyment [47,110,111]. Consequently, players with higher levels of perceived control are expected to derive greater enjoyment from the game and be more motivated to continue playing. Based on this, we propose the following hypothesis:

Hypothesis 2. Control influences perceived enjoyment positively.

3.3. Social interaction toward perceived enjoyment

Social interaction facilitates player interplay either directly between users or indirectly through mutual connections [18,101]. It is regarded as an essential component of any social game, transcending differences in players' gender and age [101,112]. Social interaction features enable players to engage in various activities, such as playing together, adding friends, competing against others, and chatting while gaming, which enhance the game's appeal and engagement [41,113,114]. Prior research has demonstrated that user interaction contributes to increased immersion [13,101], flow experiences in games [45,115,116], positive mood enhancement [40], and perceived enjoyment [116–118]. Based on these findings, we propose the following hypothesis:

Hypothesis 3. Social interaction influences perceived enjoyment positively.

3.4. Fairness toward perceived enjoyment

Fairness is defined as "an individual's perception of the distribution of rewards resulting from the investments or output-input ratios" [119]. Transactions are perceived as fair when the ratios of rewards to investments are equal, while unequal ratios lead to perceptions of unfairness.

In the freemium model, users can access basic app services for free and choose to unlock additional content, services, or items by spending money [120]. Research has shown that fairness perceptions influence attitudes toward a company, which subsequently impact purchase intentions during transitions from free to fee models [20,121]. Hamari [16] argued that the introduction of in-app purchase items can diminish players' perceptions of fairness, potentially fostering negative attitudes toward in-app purchases. Free players often perceive the victories of those who pay for virtual goods as undeserved, leading to feelings of unfairness. These perceptions can significantly reduce the enjoyment derived from playing games [53,122]. Research has shown that when players experience a mismatch between their inputs—such as time, money, skills, curiosity, concentration, and effort—and their in-game performance, they are likely to feel dissatisfied, which can result in a loss of interest and a shift to alternative games [123,124]. Similarly, players' enjoyment decreases when they observe opponents gaining unjustified advantages during gameplay [125]. Based on these insights, we propose the following hypothesis:

Hypothesis 4. Fairness influences perceived enjoyment positively.

3.5. Fairness toward envy

Envy is generally triggered when an individual desires the accomplishments or possessions of others [65]. The connection between fairness and envy is widely discussed in existing research. Envy often originates from an upward social comparison, where an individual recognizes that a similar other enjoys superiority in a particular aspect. This recognition can evoke feelings of inferiority and pose a threat to one's ego [126,127]. In a social comparison scenario, when an individual perceives others' advantages as undeserved, feelings of anger and a subjective perception of injustice may arise. These perceptions can evoke aggressive affective behavior toward the envied individual, gradually leading to malicious envy [128]. Conversely, benign envy is triggered when the envied individual is perceived to deserve their superiority. Drawing from appraisal theory [129] and the work of Van de Ven et al. [78], envy can be categorized as benign or malicious, depending on perceptions of fairness.

Malicious envy and benign envy are both intense and negative emotional states, differing primarily in the envier's perception of whether the advantages held by the envied individual are deserved. Malicious envy arises when the envier perceives the favorable position or possession of the envied individual as unjustified, whereas benign envy is evoked when the superiority is deemed fair and deserved [67,80,128]. In the context of social freemium games, two distinct situations of envy may emerge. Malicious envy is likely to be triggered when players perceive it as unfair that their peers achieve higher leaderboard positions solely through purchasing in-app items. Conversely, benign envy tends to occur when higher ranks are achieved through means perceived as fair and justified. Based on this discussion, we propose the following hypotheses:

Hypothesis 5a. Fairness influences benign envy positively.

Hypothesis 5b. Fairness influences malicious envy negatively.

3.6. Perceived enjoyment toward in-app purchase intention

Perceived enjoyment refers to the degree to which one could find the use of technology "enjoyable in its own right, apart from any performance consequences that may be anticipated" [130,131]. In the gaming context, perceived enjoyment is closely aligned with the concept of playability, which is defined as the degree to which a game can sustain enjoyment over an extended period [101].

Player enjoyment is widely recognized as the fundamental and ultimate goal of online game development [88,132], as well as a key driver of information systems (IS) adoption and continued use [14,133]. Novak et al [103] identify enjoyment as a critical dimension of the flow experience, which is strongly associated with users' online purchasing behavior. The greater the enjoyment players derive from a game, the more likely they are to continue playing and consider purchasing in-game items [27,134]. Based on this discussion, we propose the following hypothesis:

Hypothesis 6. Perceived enjoyment influences in-app purchase intention positively.

3.7. Benign and malicious envy toward in-app purchase intention

Upward social comparison is the root of envy, arising when individuals perceive that others possess desirable advantages. Despite being a negative emotion, envy is often leveraged by marketers to stimulate consumer behavior [135]. When envy is triggered, individuals tend to focus more on those they envy [72], which can drive them to make similar purchases in an effort to enhance their own status [82,83]. For example, studies have shown that envy induced by social media usage, such as surfing Facebook, can lead to conspicuous consumption [25]. Similarly, research on MacBook Pro purchases reveals that envy evoked by a social networking site (SNS) post significantly increases subsequent purchase intentions [24]. In line with this, Gunaseran and Khalid [136] highlight the link between envy and consumers' willingness to pay for a smartphone.

Envy can be classified into two types: benign envy, which motivates individuals to work harder to achieve the envied object, and malicious envy, which prompts undesirable actions aimed at undermining the envied [76,137]. These two types of envy trigger distinct motivations, leading to different behavioral outcomes. Benign envy fosters a "moving-up" motivation, encouraging individuals to improve their status, exert effort, and strive to attain the desired object. Due to its positive effects, previous studies have shown that benign envy can stimulate consumer behavior [76] and increase willingness to purchase [63,135].

Conversely, malicious envy is characterized by feelings of inferiority, hostility, and resentment directed toward the envied individual [66]. Malicious enviers frequently experience low self-esteem, anxiety, and frustration. While benign envy encourages a "leveling-up" impulse, malicious envy is more likely to evoke a "leveling-down" motivation, prompting individuals to adopt unfavorable behavior to compensate for their perceived inferiority during social comparison. Ferreira and Botelho [138] found that in the presence of malicious envy, consumers tend to associate a brand with hostile emotions and distance themselves from the envied individual, the product, and the brand. Consequently, benign envy is believed to foster a positive attitude toward a brand, whereas malicious envy is likely to generate negative experiences and perceptions associated with the brand.

In the context of freemium games, Evers et al [139] apply envy theory to explain a similar phenomenon. When players observe others gaining advantages simply by spending money, they may experience envy and feel tempted to acquire similar benefits. In freemium social games, which involve heightened levels of interaction, players are likely to be even more influenced by their partners' high ranks. Players experiencing benign envy may be motivated to purchase in-app items as a way to narrow the gap between themselves and the envied players. Conversely, malicious enviers may become frustrated with the game and, as a result, be less inclined to make in-app purchases [80,140]. Based on this discussion, we propose the following hypotheses:

Hypothesis 7a. Benign envy influences in-app purchase intention positively.

Hypothesis 7b. Malicious envy influences in-app purchase intention negatively.

Fig. 1 demonstrates the conceptual framework of the study.

4. Research methodology

4.1. Research design

This study employed a 2 × 3 × 2 between-subjects experimental design, manipulating three independent variables: fairness, skill/challenge, and control. Fairness was manipulated as fair and unfair, while control was manipulated as high or low. Following flow theory, we manipulated skill/challenge as balance, imbalance boredom, and imbalance overload [44]. Since social interaction and its related functions are embedded in social games by default, particularly in MMORPGs, this factor was measured using questionnaire items rather than being manipulated as a separate factor. In total, 12 unique scenarios were created, as summarized in Table 1.

4.2. Participants

The study involved a total of 302 respondents, with 24 to 30 participants assigned to each treatment condition. Since the designed game is relatively popular and available on the Google Play Store, participants were not limited to any nationality. The majority of respondents were university students based in Tainan City, Taiwan, where the field experiment was conducted. Previous studies of gaming behavior have frequently targeted university students due to their familiarity with gaming [6,141], which makes them well suited to provide insightful and accurate perspectives on freemium game behavior. Participation was not restricted by occupation or educational background.

To ensure the adequacy of the sample size, we conducted a post hoc power analysis using G*Power 3.1, as post hoc power assessments can serve as an additional safeguard for verifying statistical power [142]. An ANOVA analysis with fixed effects, including both main effects and interactions, was employed to evaluate whether the sample size was sufficient to detect significant main effects in the 2 × 3 × 2 between-subjects design. This design included three factors: control (two levels), skill–challenge balance (three levels), and fairness (two levels). The analysis specified a medium effect size of 0.25, a significance level of 0.05, and 12 groups. The numerator degrees of freedom for the main effects were 1 for control, 2 for skill–challenge balance, and 1 for fairness. The results showed that the sample size of 302 achieved a power of 0.991 for detecting the main effects of control and fairness, and a power of 0.978 for detecting the main effect of skill–challenge balance. Both power levels exceed the commonly accepted threshold of 0.80, ensuring that the study is well powered to identify meaningful effects.

4.3. Freemium social game selection

In this study, we collaborated with an independent developer specializing in 3D games to create a new social game (MMORPG) inspired by the popular game *Dark Souls 3*. We developed a small-scale freemium social game for the Android platform, titled *Gaia World*, which was subsequently uploaded to the Google Play Store (Appendix A).

Gaia World is a multiplayer role-playing social game featuring a variety of gaming functions that enable players to interact freely, make friends, battle monsters, and complete missions. The game employs a freemium business model, allowing users to download and play the fully

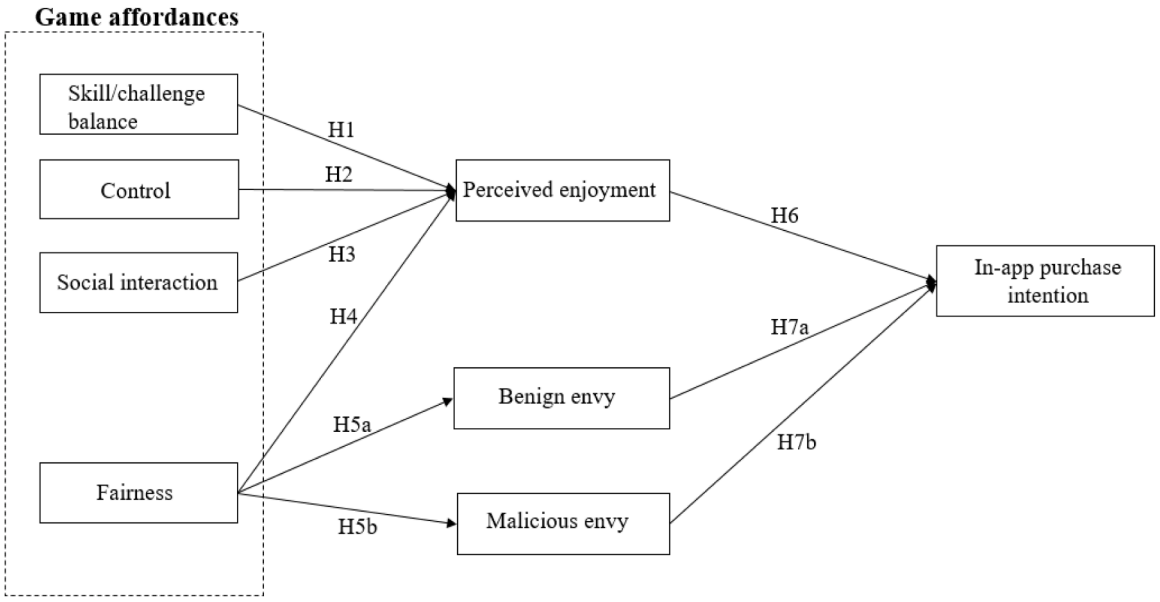

Fig. 1. Conceptual framework.

Table 1
Experimental design of the study.

		Control	Fairness	
			Fair (F)	Not Fair (NF)
Skill/ Challenge	**Balance**	**High**	Balance × H × F	Balance × H × NF
		Low	Balance × L × F	Balance × L × NF
	Boredom	**High**	Boredom × H × F	Boredom × H × NF
		Low	Boredom × L × F	Boredom × L × NF
	Overload	**High**	Overload × H × F	Overload × H × NF
		Low	Overload × L × F	Overload × L × NF

functional game for free while offering in-app transactions. These transactions include the purchase of virtual goods, such as equipment, through virtual shops (Appendix B).

We also developed a custom database to manage participants' accounts, ensuring that each participant had a unique in-game name and that their scores were securely stored. The database played a crucial role in creating a multiplayer social game (MMORPG) environment where participants could simultaneously interact and play together. The game featured various social functions, including a Quest, Inventory, Chat, Friends, an In-App Purchase Shop, and a Ranking System.

Quest refers to tasks assigned to players that must be completed to progress. To complete a quest, players needed to interact with a non-player character (NPC) by navigating to a small blue portal, transitioning to other maps using a large blue portal, and defeating all monsters in the designated area. Each player was limited to three quests per day.

During gameplay, participants who desired a better experience—such as advancing their character's status, gaining scores more quickly, or achieving victories more easily—could purchase in-app equipment from the In-App Purchase Shop using virtual currency (gems). Players could acquire gems either by completing quests or by purchasing them with real money, with 250 gems priced at US$15. The in-app purchase equipment could also be exchanged for performance scores at the end of the game, with a maximum of seven items allowed per player.

Players could check for available items as well as how many gems could be purchased using real money by clicking the "shop" button. Social features included the "Add Friend" function, which allowed players to expand their social network, and a "Chat" feature for interacting with other players. The Ranking System provided a leaderboard displaying other players' ranks, names, scores without in-app purchases, bonus scores from in-app purchases, and total scores (Appendix C), enabling participants to track their own and others' performance with ease.

4.4. Stimulus development and procedures

Our study utilized a field experiment to address the research questions. According to Babbie [143], field experiments conducted in natural settings are effective for testing controlled causal processes, as they minimize the artificiality associated with laboratory experiments, providing a more accurate reflection of users' behavior. Furthermore, the natural social context of field experiments is believed to enhance their external validity compared with laboratory experiments. An additional advantage of field experiments is their potential for accessing a larger and more diverse sample population.

We began by introducing the freemium social game (MMORPG) to participants and seeking their consent to participate in the research. Participants were provided with a consent statement to ensure the anonymity and confidentiality of their responses, which helped lessen social desirability bias [144,145].

Participants who agreed to join the experiment were instructed to download *Gaia World* from the Google Play Store and follow the steps outlined for the study. After downloading and installing the game on their Android devices, participants were required to register a new account and log in using the newly created account (Appendix D). The experiment duration ranged from 5 to 10 minutes for the pre-study phase and 5 to 15 minutes for the main study.

In this study, skills could not be manipulated, as they represented a subjective experience that varies between individuals [100,146]. Therefore, before participating in the main experiment, participants were asked to self-assess their skill levels during a pretest by playing the baseline version of the game, which excluded in-app purchase features, for 5–10 minutes (Appendix E). In the pretest, participants could view their characters in the game and were required to interact with non-player characters (NPCs) by navigating through a small blue portal, transitioning to another map via a large blue portal, and defeating all monsters. Upon defeating all monsters, participants interacted with an

NPC again to complete the mission, at which point they were automatically redirected to the pre-study questionnaire form. The self-assessment of skill levels was conducted using a 9-point scale ranging from "low skill" to "high skill," developed based on the measurement items proposed by Engeser and Rheinberg [147].

After completing the pretest for skill assessment, participants were instructed to select either "AREA 1" or "AREA 2" (Appendix E) to proceed to the main study. "AREA 1" was designated for players with low skill levels (self-evaluations ranging from 1 to 4), while "AREA 2" was intended for participants with high skill levels (self-evaluations ranging from 6 to 9). The purpose of creating these two areas was to enable the system to assign participants randomly to one of 12 scenarios featuring different versions of *Gaia World*. This random assignment technique is recognized for its ability to reduce the likelihood of confounding effects and mitigate social desirability bias [148,149].

Participants in the main study were instructed to play the game for 5 to 15 minutes. Their primary task was to defeat all monsters and the boss to complete the quest. Players were limited to completing three quests per day during the main study. After finishing the quest, participants were asked to open the ranking list to view their scores alongside those of the top five players, which were system-generated. The ranking list displayed the players' rank, name, equipment used, score without in-app purchases, bonus score from in-app purchases, and total score (Appendix C). Participants were asked to compare their scores with those of the top five players, specifically examining scores gained from quest completion and equipment scores. Following this, they completed an online questionnaire, which included a manipulation check and the main study questions. For the in-app purchase intention factor, participants were instructed to consider all the possibilities in the question items (Appendix M) of acquiring in-game content and respond to the respective question items by rating their likelihood of making in-app purchases on a 7-point Likert scale.

4.5. Manipulation check

4.5.1. Manipulation for skill–challenge balance

Skill is defined as a player's confidence in their ability to solve gaming challenges [100,146]. As discussed above, since skill is a subjective perception, it could not be manipulated but was instead self-assessed by the participants. To facilitate this evaluation, participants were instructed to play a baseline version of the game to gauge their skill level. Based on their self-assessment, participants selected the appropriate area: AREA 1 for low-level skills or AREA 2 for high-level skills.

Challenge refers to the degree of difficulty a player perceives when attempting to complete gaming tasks [100,146]. In high-challenge conditions, users were exposed to attack areas (depicted as blue dots), which is called "skill" or "magic" in the game (Appendix F). Conversely, in low-challenge conditions, no attack areas were present, making it easier and faster to defeat monsters (Appendix G).

Skill–challenge balance was manipulated across three conditions: balance (high skill/high challenge or low skill/low challenge), imbalance boredom (high skill, low challenge), and imbalance overload (low skill, high challenge). Participants in the main study were asked to experience the game under one of these conditions and then complete a questionnaire to determine if there was any difference between balance, boredom, and overload conditions. The questionnaire items were adapted from Jackson and Marsh [150] and employed a 7-point Likert scale for responses.

4.5.2. Manipulation for control

Control is defined as an individual's perception of their ability to manage a specific interaction [102]. In the gaming context, control also encompasses the user's influence, guiding capabilities, and dominance within the game environment [103]. Users who perceive greater control experience a heightened sense of freedom and the ability to act as they desire. In MMORPGs, control is manifested through features such as shortcut bars, macro commands, and controllers [109]. These features simplify character management and facilitate the completion of various tasks, enhancing the overall gaming experience [13,109].

In this study, control was manipulated through two conditions: high control and low control. In the high control condition, a virtual controller was provided, enabling players to control their characters' movements with greater ease (Appendix H). Conversely, in the low control condition, no virtual controller was available, and players could only manage their characters by double-tapping them during gameplay (Appendix I).

Participants were instructed to experience the game under their assigned condition and then complete a questionnaire to assess whether the perceived control level was high or low. Responses were collected using a 7-point Likert scale.

4.5.3. Manipulation for fairness

Fairness is defined as "an individual's perception of the distribution of rewards resulting from investments or output–input ratios" [119]. In this study, input refers to the methods players use to complete quests and earn points, which may include their gaming skills or the money spent to purchase virtual currency (gems). Gems can be exchanged for in-app items and bonus scores. Output is defined as the final points obtained by each player.

As outlined earlier, players' final scores could be earned through quest completion or by exchanging in-app purchase items. Completing the quest awarded 20 points for defeating each monster and 80 points for defeating the boss. To acquire in-app purchase items, players could buy virtual currency (gems) using real money. In this study, the cost of one in-app purchase was set at 250 gems, which could be purchased for 15 USD. Alternatively, players could earn virtual currency (gems) by leveraging their skills to complete quests.

According to Lind and Tyler [151], fairness can be evaluated based on the consequences of an event (distributive justice) or the procedures through which the outcome is achieved (procedural justice). The process of obtaining an outcome can be used to manipulate perceived fairness, as procedural information is typically presented before outcome information.

Based on the discussion, we manipulated game fairness by asking respondents to assess the game policy regarding the output/input ratio. It was expected that participants' perceptions of game fairness would be influenced directly by the perceived fairness of the policy, which was presented prior to the outcome information. Specifically, participants evaluated fairness by comparing the expected final points (output) derived from two methods (input): completing the quest using their skills and exchanging in-app purchases for extra points.

We developed various game policy scenarios and asked participants to categorize the content as either fair or unfair (Appendix J). To enhance clarity, we provided screenshots and brief explanations for each policy scenario. Pretest results from 25 respondents indicated that for the fair condition, 72 % selected the game policy stating "One in-app purchase item can be exchanged for 300 extra points in the final score. Gems can also be earned by completing quests (40 gems per quest)" (Appendix K). Under this policy, with the fixed rule allowing only three quests to be completed per day, players could earn a maximum of 120 gems daily. As previously established, the fixed condition required 250 gems to purchase one in-app item. Therefore, in this scenario, players would need at least three days to earn enough gems for one in-app purchase item.

For the unfair condition, 92 % of respondents identified the game policy as unfair when it stated "One in-app purchase item can give you 300 additional points in your score. Gems can also be earned by completing quests (5 gems per quest)" (Appendix L). Under this policy, with the limitation of three quests per day, players could earn a maximum of 15 gems daily, requiring 17 days to accumulate enough gems for one in-app purchase item.

In the main study, after completing their quest, participants were asked to open the ranking list to view their own scores and compare them with those of other players, as well as to see the equipment used by the top players. The ranking list displayed each player's rank, name, score without in-app purchases, bonus score from in-app purchases, and total score.

The top five players on the ranking list were system-generated. These players all had in-app purchase equipment and scores that varied depending on whether the scenario was fair (see Appendix 8) or unfair (see Appendix 9). Participants were instructed to compare their own scores and equipment with those of the top five players according to their assigned scenarios. Afterward, they completed a questionnaire assessing their perception of fairness in the freemium game, using a 7-point Likert scale.

4.6. Measurements

This study utilized a 42-item survey questionnaire to collect data from the sample respondents. In our study, skill–challenge balance refers to a game's capacity to maintain an equilibrium between its level of difficulty and the player's skillsets. The measurement items for skill–-challenge balance were designed to capture the alignment between the challenges presented in a game and the player's skills. To measure skill–challenge balance affordance, we adopted measurement items from Jackson and Marsh [150] that aligned closely with those used in previous MMORPG studies [27,28]. For control, we followed an approach similar to [46,47], defining control affordance as the integration of features that enable players to engage with and manipulate game objects and properties through available physical control mechanisms. The items for control were adopted from Lowry et al [152], which effectively capture the diverse control possibilities afforded to players during gameplay. Social interaction, referred to as social affordance in the study in [40], describes the extent to which MMOGs facilitate player engagement with others within the virtual world. We adopted the measurement items for social interaction from Su et al [101]. reflecting the game capabilities that enable players to participate in both communicative and cooperative activities within the game, aligning with the operationalization in the study by [40]. Consistent with the conceptualization of game fairness in the studies [28], the measurement of fairness was derived from Crusius and Lange [77] and Ding and Lii [153]. Our measurement items for game fairness reflect the concept of "justice"—the perception that the gaming environment is equitable and conducive to achieving player objectives [52]. The measurement items of perceived enjoyment were adapted from Lowry et al. [152]. Items for benign envy and malicious envy were adopted from Crusius and Lange [77], and the in-app purchase intention construct was measured using items adapted from Li et al [154]. All survey items were rated on 7-point Likert scales, ranging from "strongly disagree" to "strongly agree" (Appendix M). Additionally, gender, age, education, income, frequency of use, years of usage, and nationality were included as control variables, following the frameworks of previous studies [155,156].

5. Data analysis and results

5.1. Characteristics of the respondents

Table 2 presents the demographic characteristics of the participants. Male respondents constituted the majority, making up 66.2 % of the sample, which is double the proportion of females (33.8 %). Most respondents were aged 18 to 34 years (95 %), with the majority originating from Indonesia (84 %) and holding a bachelor's degree (53 %). Furthermore, 42 % of participants reported having over four years of experience with freemium social games, while 60 % played for less than 2 hours per day.

Table 2
Characteristics of respondents.

Respondent Information	Category	Distribution	Percentage (%)
Gender	Male	200	66.20 %
	Female	102	33.80 %
Age	Under 18 years old	2	0.70 %
	18–24 years old	141	46.70 %
	25–34 years old	145	48.00 %
	35–44 years old	12	4.00 %
	45–54 years old	1	0.30 %
	55 years old and above	1	0.30 %
Nationality	Indonesia	254	84.10 %
	Vietnam	19	6.30 %
	Thailand	13	4.30 %
	Taiwan	7	2.30 %
	Others	9	3 %
Education	High School or Below	36	11.90 %
	Bachelor's Degree	159	52.60 %
	Master's Degree	99	32.80 %
	Doctoral Degree	8	2.60 %
Income	Less than U $200	111	36.8 %
	US$200–US$300	69	22.8 %
	US$300–US$400	53	17.5 %
	US$400–US$500	26	8.6 %
	More than US$500	43	14.2 %
Freemium social games experience	Less than 1 year	105	34.80 %
	1 year	18	6.00 %
	2 years	33	10.90 %
	3 years	15	5.00 %
	4 years	4	1.30 %
	More than 4 years	127	42.10 %
Frequency of playing	Less than 1 hours/day	107	35.40 %
	1–2 hours/day	76	25.20 %
	2–4 hours/day	58	19.20 %
	4–6 hours/day	25	8.30 %
	6–8 hours/day	18	6.00 %
	More than 8 hours/day	18	6.00 %

5.2. Manipulation check results

To evaluate the effectiveness of the manipulations, ANOVAs were conducted using IBM SPSS 20 to test the overall manipulation checks for skill/challenge, control, and fairness. The results of univariate analysis of variance revealed significant differences under all conditions. For skill/challenge, there were significant differences between balance, boredom, and overload conditions ($M_{balance}$=5.02, $M_{boredom}$=5.41, $M_{overload}$=3.87, F-value=48.620, p-value < 0.001, $\eta_p 2 = 0.245$). For control, significant differences were observed between high and low control conditions (M_{high}=4.90, M_{low}=3.49, F-value=82.820, p-value < 0.001, $\eta_p 2 = 0.216$). Similarly, the fairness manipulation showed a significant difference between fair and unfair conditions (M_{high}=5.47, M_{low}=3.76, F-value=162.330, p-value < 0.001, $\eta_p 2 = 0.351$). In summary, all mean scores differed significantly in the expected direction, confirming the success of the experimental manipulations.

5.3. Measurement properties

After analyzing the manipulated variables using SPSS 20, we utilized AMOS 20 to test the measurement model. Table 3 presents the means, standard deviations, reliability scores (Cronbach's a and composite reliability), and average variance extracted (AVE) for all factors, along with the correlations among them. The quality of the measurements for all factors was thoroughly assessed. First, both Cronbach's a and composite reliability values for each factor exceeded the recommended thresholds, confirming the reliability of the measurements. Second, all factor loadings were above the recommended minimum value of 0.60,

Table 3
Descriptive statistics, correlation coefficients, reliability, and validity.

	Mean	SD	CA	CR	AVE	IAPI	PF	SC	CL	SI	PE	ME	BE
IAPI	4.26	1.6645	0.917	0.921	0.796	0.892							
PF	4.62	1.4413	0.895	0.897	0.687	0.379	0.829						
SC	4.77	1.3227	0.887	0.889	0.667	0.096	0.096	0.817					
CL	4.2	1.5118	0.879	0.88	0.711	0.115	0.181	0.212	0.843				
SI	5.36	1.1096	0.822	0.828	0.618	0.386	0.22	0.213	0.173	0.786			
PE	4.79	1.3472	0.909	0.912	0.777	0.391	0.337	0.431	0.349	0.414	0.882		
ME	3.42	1.6194	0.929	0.93	0.655	-0.012	-0.29	0.08	0.023	0.123	0.009	0.809	
BE	4.71	1.4575	0.926	0.925	0.673	0.438	0.46	0.29	0.058	0.204	0.393	0.021	0.821

Notes: CA: Cronbach's a; CR: composite reliability; AVE: average variance extracted; IAPI: In-App Purchase Intention; SC: Skill/Challenge balance; CL: Control; SI: Social Interaction; PE: Perceived Enjoyment; ME: Malicious Envy; BE: Benign Envy.

except for CL3, CL4, SI4, PE3, PE4, PE6, and PF5, which were subsequently removed from the model (Appendix M).

The square root of the AVE for each factor (diagonal elements) exceeded the correlations between the factor and other factors (off-diagonal elements), indicating high discriminant validity. Additionally, the CFA results demonstrated good fit indices for the measurement model, with a $\chi 2/df$ of 1.67, a GFI of 0.871, an AGFI of 0.843, an NFI of 0.899, and a CFI of 0.957. The root mean squared residual (RMR) and the root mean squared error of approximation (RMSEA) were also acceptable, at 0.047 and 0.123, respectively. These results confirm the parsimony of the model, along with high reliability and validity. Overall, the variables in the measurement model demonstrate adequate reliability and validity.

5.4. Tests of hypotheses

The structural model was tested using AMOS 20 to evaluate path significance levels. In this experiment, all conditions included identical survey items. As summarized in Table 4, the overall model fit was acceptable with $\backslash\chi 2$ of 800.5, df of 462, $\chi 2/df$ value of 1.747, GFI value of 0.861, AGFI value of 0.835, NFI value of 0.891, CFI value of 0.950, RMR value of 0.205, and RMSEA of 0.050 with p-value at 0.000.

As anticipated, skill–challenge balance, control, and social interaction all influenced perceived enjoyment positively (ß = 0.314, $p < 0.001$; ß = 0.199, $p < 0.001$; ß = 0.269, $p < 0.001$), supporting Hypotheses 1, 2, and 3. Fairness had a positive effect on both perceived enjoyment (ß = 0.222, $p < 0.001$) and benign envy (ß = 0.443, $p < 0.001$) while affecting malicious envy negatively (ß = –0.275, $p < 0.001$), thereby supporting Hypotheses 4, 5a, and 5b.

Additionally, both perceived enjoyment and benign envy significantly influenced in-app purchase intention (ß = 0.269, $p < 0.001$; ß = 0.343, $p < 0.001$). However, malicious envy showed no significant relationship with in-app purchase intention (ß = –0.018, $p > 0.05$), supporting Hypotheses 6 and 7a but not Hypothesis 7b. The path coefficients are summarized in Figure 2.

Table 4
Results of structural equation model.

Hypothesis/Path		Std. Coefficient	*t*-value	Model Fit Statistic
H1	Skill/challenge balance → Perceived Enjoyment	0.314***	5.707	$\chi 2 = 800.5$, df = 462; $\chi 2/df = 1.747$; $p = 0.000$; GFI = 0.861; AGFI = 0.835; NFI = 0.891; CFI = 0.950; RMR = 0.205; RMSEA = 0.050
H2	Control → Perceived Enjoyment	0.199***	3.647	
H3	Social Interaction → Perceived Enjoyment	0.269***	4.678	
H4	Fairness → Perceived Enjoyment	0.222***	4.136	
H5a	Fairness → Benign Envy	0.443***	7.492	
H5b	Fairness → Malicious Envy	–0.275***	–4.448	
H6	Perceived Enjoyment → In-App Purchase Intention	0.269***	4.688	
H7a	Benign Envy → In-App Purchase Intention	0.343***	5.795	
H7b	Malicious Envy → In-App Purchase Intention	–0.018	–0.334	

5.5. Post hoc test

Post hoc analyses were conducted to examine the effect of imbalance conditions on perceived enjoyment. As shown in Table 5, the results indicated that the balance condition enhanced players' enjoyment, aligning with previous findings [27]. Interestingly, imbalance conditions were also capable of generating enjoyment, with the imbalance–boredom condition resulting in significantly higher enjoyment than the imbalance–overload condition (mean difference = 0.9279, $p < 0.001$). Perceived enjoyment under the balance condition was not significantly different from that under the imbalance–boredom condition ($p > 0.05$). However, enjoyment under the balance condition was significantly higher than under the imbalance–overload condition (mean difference = 0.6045, $p < 0.001$).

5.6. Control variables

This study included several control variables, such as gender, age, education, income, freemium social game experience, frequency of gameplay, and nationality. The purpose of analyzing these control variables was to determine whether they had any impact on the research model. The results confirmed that none of the control variables influenced either the independent or dependent variables significantly.

6. Discussion

This study has sought to understand how game affordances influence users' experiences in freemium social games, specifically focusing on perceived enjoyment and envy, and how these positive and negative experiences shape users' in-app purchase intentions. First, the results revealed that skill–challenge balance influences perceived enjoyment significantly, consistent with the findings of Wang et al [27], who argue that perceived enjoyment is maximized when challenge and skill are well matched. Interestingly, imbalance conditions (boredom and overload) are also found to evoke enjoyment. Post hoc analysis further indicates that there is no significant difference in enjoyment between balance and imbalance-boredom conditions. Additionally, both balance and imbalance-boredom conditions significantly generate greater enjoyment than the imbalance-overload condition. These findings are consistent with [157], which demonstrates that when players face low

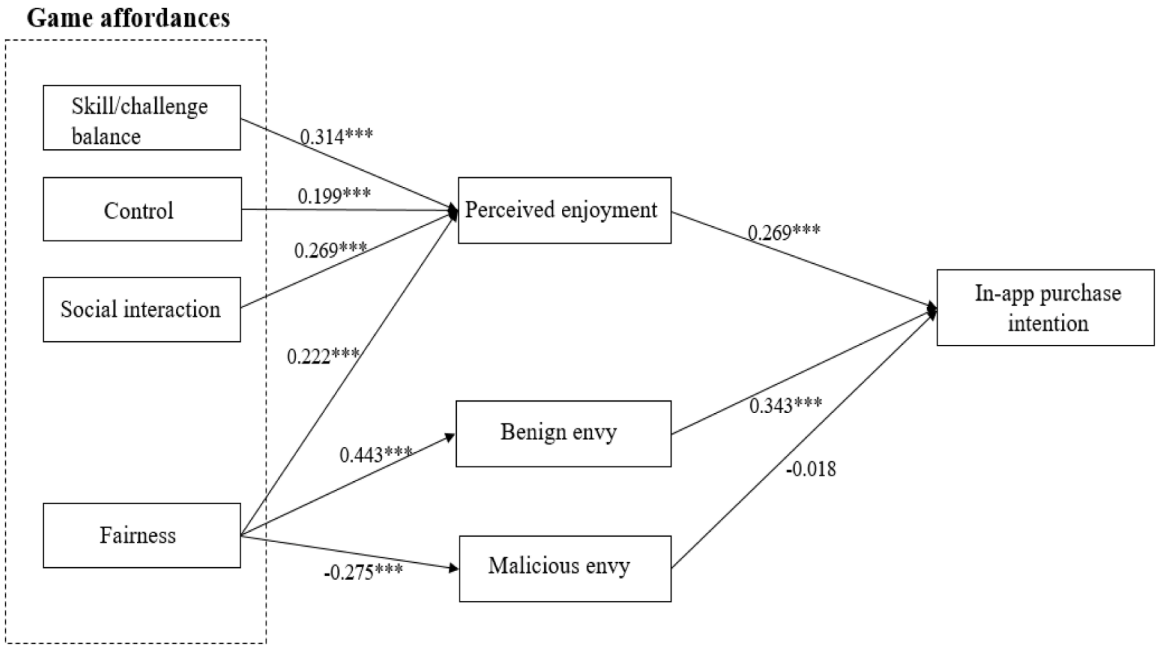

Fig. 2. Results of structural equation model (SEM) analysis. *Notes:* Significance of path: * significant at level 0.05, ** significant at level 0.01, *** significant at level 0.001. All factor loadings are significant at level 0.001 and are not put in the asterisks.

Table 5
Results of post hoc analysis.

Dependent Variable	Manipulated Independent Variables		Mean Difference	Std. Error	*p*-value
Perceived Enjoyment	Skill/Challenge Imbalance (Overload)	Imbalance (Boredom)	–0.9279*	0.1769	0.000
		Balance	–0.6045*	0.1769	0.001
	Imbalance (Boredom)	Imbalance (Overload)	0.9279*	0.1769	0.000
		Balance	0.3234	0.1764	0.068
	Balance	Imbalance (Overload)	0.6045*	0.1769	0.001
		Imbalance (Boredom)	–0.3234	0.1764	0.068

levels of challenge, the likelihood of successfully defeating enemies increases, while the probability of failure decreases, enhancing players' enjoyment compared with highly challenging scenarios. Similarly, in this study, the imbalance-boredom condition allows players to complete quests more easily, reduces the risk of failure, and is more likely to elicit enjoyment.

Second, control is confirmed to play a crucial role in enabling players to interact with the game effectively. The results indicate that high control enhances the gaming experience, leading to greater enjoyment. The availability of virtual controllers provides players with greater freedom to make choices within the game and achieve their goals more easily. These advantages undeniably contribute to players' enjoyment during gameplay. This finding aligns with existing evidence suggesting that a sense of control is essential for understanding why individuals enjoy games and is a key factor in their success within the game [47,110,158].

Third, social interaction within the game is found to influence perceived enjoyment positively, consistent with findings from previous studies [40,101,116]. As social interaction is a fundamental human need, even brief interactions can make the gaming experience more engaging. Technologies that enhance social interactions further increase levels of enjoyment and immersion [159].

Fourth, the study yields consistent findings with extant literature on envy, which suggests that players experience benign envy when they perceive fair treatment, while malicious envy arises from perceptions of unfairness [78,136,160,161].

Another critical finding is that fairness can significantly induce players' enjoyment of the game. Lin and Sun [53] suggest that dissenters who prioritize the hedonic aspects of gaming often criticize game designers for deliberately creating a stark disparity between paying and non-paying players. Specifically, non-paying players are often required to engage with a less enjoyable version of the game. Moreover, decisions regarding in-app purchases can influence player rankings directly, as in-app purchase items provide advantages that make it challenging for free players to compete with paying players' scores [20]. In such scenarios, free players may perceive unfair treatment, which can impact their enjoyment of the game negatively. These findings accord with previous discussions on the implications of the freemium business model in studies such as [27,28].

The significant relationship between perceived enjoyment and in-app purchase intention is in line with findings from prior research [134,141,162,163]. Participants who find freemium social games highly compelling and immensely enjoyable are more likely to seek additional gratification from the game, which motivates them to invest more time and money in purchasing virtual items [27].

Additionally, benign envy is found to strengthen in-app purchase intention significantly. In the context of freemium social games, players experiencing benign envy toward others may attempt to reduce the gap by striving to obtain what others have, ultimately leading to in-app purchase intention. These findings are consistent with previous studies indicating that benign envy drives individuals to purchase the envied product [80,164–166].

Last, the relationship between malicious envy and in-app purchase intention is found to be nonsignificant. This result may be attributed to the tendency of players experiencing malicious envy to avoid acknowledging such feelings, either to themselves or others, in an effort to preserve their self-images [66,78,83]. Additionally, malicious envy is often linked to low self-esteem, anxiety, and frustration. While benign envy motivates individuals to "level up," malicious envy is more likely to evoke a "leveling down" motivation, leading individuals to adopt unfavorable behavior as a means of compensating for perceived inferiority during social comparison. For instance, Parks et al [167]. find that

individuals who fail to acknowledge the advantages of others tend to exhibit less cooperative behavior with co-workers. Similarly, in the context of social networking sites (SNS), individuals experiencing unfavorable emotions triggered by malicious envy may reduce their SNS usage intensity, deactivate their accounts, or switch to alternative platforms [168–170]. In the context of freemium games, players unwilling to pay for virtual items may avoid interacting with those who use them, ultimately leading to disengagement from the game [171]. Maliciously envious players may decide to stop playing the game altogether and switch to an alternative as a way of escaping undesirable situations. Moreover, while existing evidence clearly demonstrates the stimulating role of benign envy on in-app purchase intentions, the impact of malicious envy remains inconclusive [136].

7. Theoretical and managerial implications

7.1. Theoretical implications

This study contributes to the literature in several meaningful ways. It advances the field of freemium social games by integrating affordance theory and social comparison theory into a comprehensive model that elucidates the psychological factors influencing purchase intentions. The findings provide valuable insights into how game affordances and social comparison dynamics enhance our understanding of user behavior. By extending affordance theory to the context of freemium social games, this study enriches its application and highlights its potential for future research on purchase intentions. Specifically, it sheds light on how the action possibilities offered by technological affordances are intricately connected to users' intentions to make in-app purchases.

Our analysis uncovered the mechanisms underlying the relationship between technology affordances and usage behavior in the context of freemium social games. We identified key affordances and examined how they influence players' perceived enjoyment and envy, which subsequently shape their intention to make in-app purchases. These findings provide valuable insights into the ways game affordances impact in-app purchasing decisions. To the best of our knowledge, this study is among the first to integrate game affordances, player experiences, and in-game purchasing behavior into a unified theoretical framework. While much of the existing research focuses on the link between affordances and positive behavior, little is known about how affordances contribute to either positive or negative emotions. Our contribution to the affordances literature highlights their critical role in explaining users' emotional responses and subsequent behavior.

This research also makes a significant contribution by presenting a theoretical model that elucidates users' perceived enjoyment within the unique context of freemium social games. While previous studies predominantly emphasize the balance condition as a determinant of enjoyment, this study reveals that imbalance conditions of skill and challenge—specifically boredom and overload—also have a substantial impact. These findings enhance the existing literature on perceived enjoyment and provide deeper insights into the critical elements that contribute to the hedonic value of freemium social games.

This study applies Festinger [56]. 's social comparison theory and extends its application to the novel context of freemium social games. Since social comparisons are actively encouraged by game developers, users increasingly experience envy, making it essential to understand its impact on purchase intentions and other consequent behavior. We have integrated not only the antecedents and consequences of envy but also its different types into the research framework, which makes the study more comprehensive. By distinguishing between benign envy and malicious envy, this study elucidates their distinct impacts and underlying drivers. These findings pave the way for future research examining emotional responses related to in-app purchases. Furthermore, this study contributes to a deeper understanding of envy, an emotion often regarded negatively. The insights can help game designers recognize the influence of envy and leverage it to enhance players' well-being and engagement.

In addition, this study is the first to investigate both the positive and negative aspects of users' experiences in freemium social games simultaneously. While research on the negative effects of these games remains limited, this study attempts to address this gap by identifying envy as a negative experience associated with fairness perception in freemium social games. This emotion is triggered as users engage with the game and make social comparisons.

As hypothesized, fairness is significantly associated with envy, which, in turn, impacts in-app purchase intention. These findings enrich the fairness literature by applying the concept of envy to explain fairness perception in the relatively under-researched context of freemium social games. The results highlight the critical role of socio-psychological factors in examining in-app purchase intentions and potentially other systems that involve social interaction. Broadly, these findings contribute to the study of purchasing intentions in freemium games and extend the scope of research within the field of information systems (IS).

Regarding methodology, this study is among the few to employ field experiments to investigate the influence of emotions on users' purchase intentions. It is also the first empirical study to manipulate fairness within a realistic research setting, effectively demonstrating how fairness perceptions drive envy in real-world scenarios. By doing so, this research diversifies the methodologies used in the information systems (IS) field and enhances the applicability of its findings.

7.2. Managerial implications

From a managerial perspective, this research could benefit game developers and marketers. The findings indicate that gamers' perceived skill–challenge balance and imbalance boredom enhance perceived enjoyment, which subsequently increases in-app purchase intentions. Based on these results, we recommend that game providers introduce tasks with varying levels of challenge to accommodate the diverse skill levels of players. To further generate a sense of enjoyment, game providers could offer free demonstrations with limited scenarios or provide guidance materials, such as tutorial videos. These resources would allow potential players to practice gaming skills, become familiar with the game mechanics, and enjoy the benefits associated with joining virtual communities. Additionally, since control was found to strengthen perceived enjoyment significantly, game providers are encouraged to offer players more options for virtual controllers to enhance their gameplay experience.

An important finding of this study is the significant role of interaction among players in enhancing perceived enjoyment, which subsequently influences in-app purchase intentions. Based on this insight, we recommend that managers foster positive user emotions by providing functionalities or services designed to strengthen relationships among players. Features that encourage interaction—such as team-based tasks, chat systems, or community events—can help players build connections within the gaming community. Gamers who interact frequently with others are more likely to achieve mutual goals, learn new skills from their peers, and experience increased enjoyment, ultimately driving their engagement and purchase intentions.

It is crucial for developers to design games in a way that fosters fairness, enhancing players' enjoyment and benign envy while minimizing the risk of malicious envy. Ensuring fairness reduces the likelihood of harmful behaviors by envious players aimed at undermining others. From the perspective of game companies, it is essential to design game policies related to the distribution of game performance carefully—such as rewards, bonus scores, and recognition—to ensure equitable treatment of both paid and free players. Additionally, game developers should clearly communicate the benefits of in-app purchase items, such as the specific enhancements or powers they provide. Furthermore, service providers could detail the amount of in-app currency that can be earned through completing quests. Providing such transparent information enables users to better assess the game's

fairness, potentially encouraging in-app purchases by fostering benign envy and an enjoyable gaming experience.

Our findings indicate that users experiencing benign envy are likely to exhibit higher in-app purchase intentions as a means of improving their ranking positions. This suggests that strategic approaches can be employed to leverage social comparison proactively to encourage purchasing behavior. Additionally, the study highlights the importance of emphasizing social structures to enhance the effectiveness of boosting sales. To this end, game providers should focus on increasing the visibility of users' ranking positions and in-game achievements. By making these metrics more prominent, players may be motivated to pursue higher positions through in-app purchases, thereby driving engagement and revenue.

This study holds value not only for game developers and marketers but also for game players. In freemium social games, players may compare themselves with others who own in-app purchase features, potentially triggering negative emotions such as malicious envy. Since maliciously envious players are less likely to make purchases, as indicated by the study findings, users can safeguard themselves emotionally by adopting the mindset of a social player—one who prioritizes building friendships and fostering healthy competition within the gaming community. Social players are generally willing to pay for a limited number of features in freemium games but tend to spend less than achievers. Achievers are more competitive and focused on gaining recognition, which drives their willingness to spend more to secure top leaderboard positions. As social players, they can still exchange value with the gaming community, contribute positively, and even form real-life connections with other players. Therefore, users should decide consciously whether to play as achievers or social players by aligning their gaming approaches with their personal goals and motivations.

8. Research limitations and future research

This study is not without its limitations. First, the research focuses exclusively on the effects of envy and flow experience within a specific type of freemium social game—massively multiplayer online role-playing games (MMORPGs). Hence, the findings may not generalize to other types of freemium social games, such as social network games (SNGs), massively multiplayer first-person shooters (MMOFPSs), multiplayer online battle arenas (MOBAs), or non-social games. Future research could address this limitation by exploring other freemium game types and identifying the key factors that influence in-app purchases within these contexts.

This study focuses primarily on perceived enjoyment as a positive emotion and envy as a negative emotion. Future research could expand on this by exploring other positive emotions, such as love or happiness, as well as negative emotions, such as sadness or anger. Additionally, the findings reveal that imbalance-boredom enhances players' enjoyment more significantly than the imbalance-overload condition. This suggests that investigating the effects of imbalance conditions on player experiences could be a promising avenue for future research.

The relationship between fairness and envy may be influenced by individual factors such as personality traits (e.g., dispositional envy, the Big Five personality traits) and interpersonal closeness (e.g., tie strength). These factors could, in turn, affect players' intentions to purchase in-app game items. By focusing solely on situational envy, this study may have overlooked the potential impact of dispositional envy on players' behavioral intentions. Future research could consider exploring both types of envy to offer a more comprehensive understanding of their effects in the context of freemium games. Fang et al [29]. find that tie strength among players drives individual paying behavior in a social gaming environment. Moreover, beyond the social and gaming variables identified in this study as affecting in-app purchase intentions, other game-related factors—such as game ratings and the availability of free alternative games—may also exert an impact on the dependent variables. Hence, it would be more thorough to consider the role of these factors in shaping user behavior and purchase intentions in future research.

Furthermore, fairness is typically conceptualized across three dimensions: interactional fairness, procedural fairness, and distributive fairness [172]. Due to limitations in the experimental design, we focused only on distributive fairness, operationalized through the design of game distribution, particularly the benefits provided by paid versions relative to the effort expended. Future research should instead measure fairness in its multidimensionality to come up with a more comprehensive understanding and yield more insightful results.

This research utilized the Google Play Store to distribute the experimental game. However, due to constraints on time and financial resources, the game was not fully optimized, and certain errors were not resolved for all Android smartphone models. These limitations may have affected participants' perceptions and gaming experiences, potentially leading some to be unwilling to download the game via the Play Store. To address these issues, future research should focus on developing a more refined and fully optimized version of the freemium social game to ensure a seamless user experience and encourage full participation from respondents.

Another limitation of this study concerns the manipulation of skill/challenge imbalance (i.e., boredom and overload). In the balance condition, participants with both high and low skill levels were randomly assigned to different scenarios. However, in the boredom and overload conditions, only high-skilled and low-skilled participants were included, respectively. To improve the randomization of respondents in future research, a technical algorithm could be employed to generate random integers. This approach would help mitigate the impact of confounding effects by ensuring that treatments and respondents in the control group differ solely due to chance when exposed to random game scenarios.

9. Conclusions

In summary, this study has investigated the impact of game affordances on users' in-app purchase intentions within the context of freemium social games. Five out of six hypotheses were supported. Specifically, perceived enjoyment and benign envy were identified as significant factors that strengthen players' in-app purchase intentions. These experiences are triggered by various game affordances, including skill–challenge balance, control, social interaction, and fairness. The novel findings of this research contribute to the existing literature on perceived enjoyment and envy by elucidating the relationship between different levels of skill-challenge balance and enjoyment, as well as by distinguishing the impacts of benign and malicious envy on users' purchasing intentions. These insights provide practical implications for game developers and other businesses employing similar freemium models, offering strategies to encourage users' willingness to make in-app purchases.

Contributions

Each author declares substantial contributions through the following:

(1) the conception and design of the study, or acquisition of data, or analysis and interpretation of data, (2) drafting the article or revising it critically for important intellectual content,

Please indicate for each author the author contributions in the text field below. Signatures are not required.

The first author Dr Trang Nguyen helps to build the research model, check the data analysis, write and revise the manuscript. The second author (corresponding author) Professor Jengchung Victor Chen helps to form research topic, build the research model, supervise research progress, revise paper, seek for funding support, and investigation. The third author Mr Julius Wonodihardjo helps to initiate the research idea, collecting data, and data analysis.

CRediT authorship contribution statement

Trang Nguyen: Writing – original draft, Validation, Supervision, Formal analysis, Conceptualization. **Jengchung Victor Chen:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Julius Wonodihardjo:** Writing – original draft, Visualization, Software, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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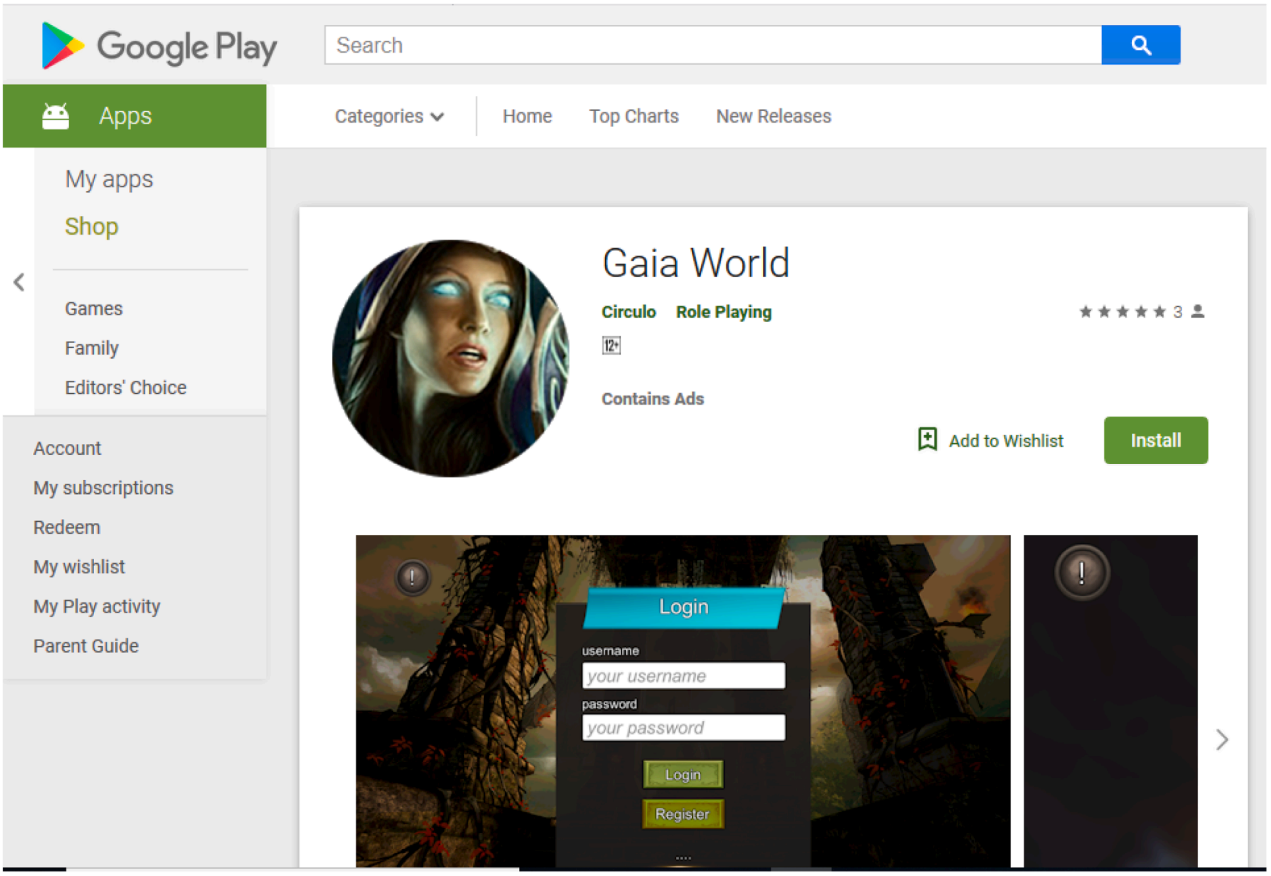
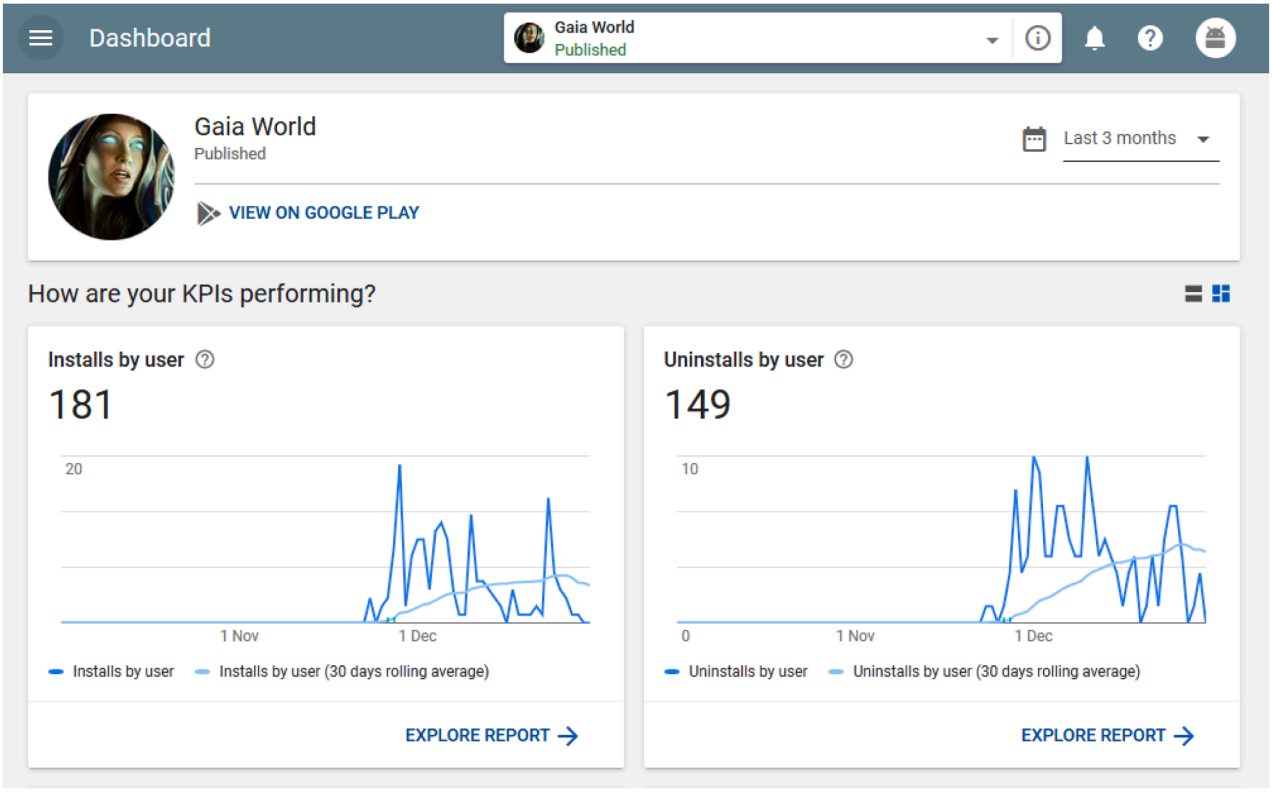
Appendix A. Google Play Store

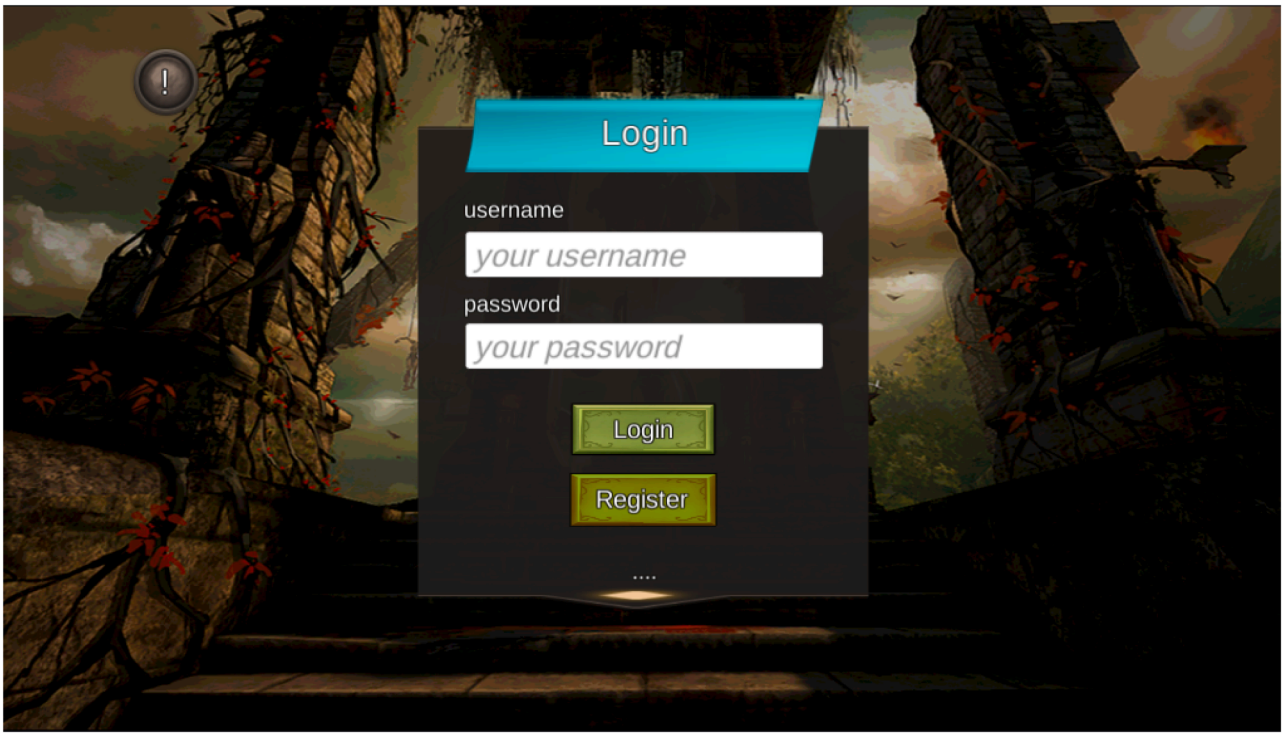

Google Play
Search
Apps
Categories
Home
Top Charts
New Releases
My apps
Shop
Games
Family
Editors' Choice
Account
My subscriptions
Redeem
My wishlist
My Play activity
Parent Guide
Gaia World
Circulo
Role Playing
3
Contains Ads
Add to Wishlist
Install
Login
username
your username
password
your password
Login
Register


Dashboard
Gaia World
Published
Gaia World
Published
Last 3 months
VIEW ON GOOGLE PLAY
How are your KPIs performing?
Installs by user
181
20
1 Nov
1 Dec
Installs by user
Installs by user (30 days rolling average)
EXPLORE REPORT
Uninstalls by user
149
10
0
1 Nov
1 Dec
Uninstalls by user
Uninstalls by user (30 days rolling average)
EXPLORE REPORT

Appendix B. Gaia World's In-App Purchase Shop

Appendix C. Ranking List

Appendix D. Gaia World's Interface



Appendix E. Gaia World (Assessment Interface)

Available Quest
Kill 1 Orc Leader
Receive Quest
Kill 1 Orc Leader
Find and kill 1 Orc Leader. I hear from my army that It's in Boss Arena
Orc Leader : 0 / 1
+0 Gem(s), +0 Exp, +0 Quest Score
Accept

HP 46%
SP 100%
Orc Leader
INVENTORY
QUEST
IS BROADCAST
schro Connected
schro: [None]
schro Connected
schro: [None]
schro Connected
Enter text...
SEND
schro
Kill this guy,
then you can finish your quest
Defense
Attack
Roll
Unlock Target
PeerCreated

WARRIOR
Assess
Area 1
Area 2

Appendix F. Gaia World (High Challenge Interface)

Appendix G. Gaia World (Low Challenge Interface)

Appendix H. Gaia World (High Control Interface)

Appendix I. Gaia World (Low Control Interface)

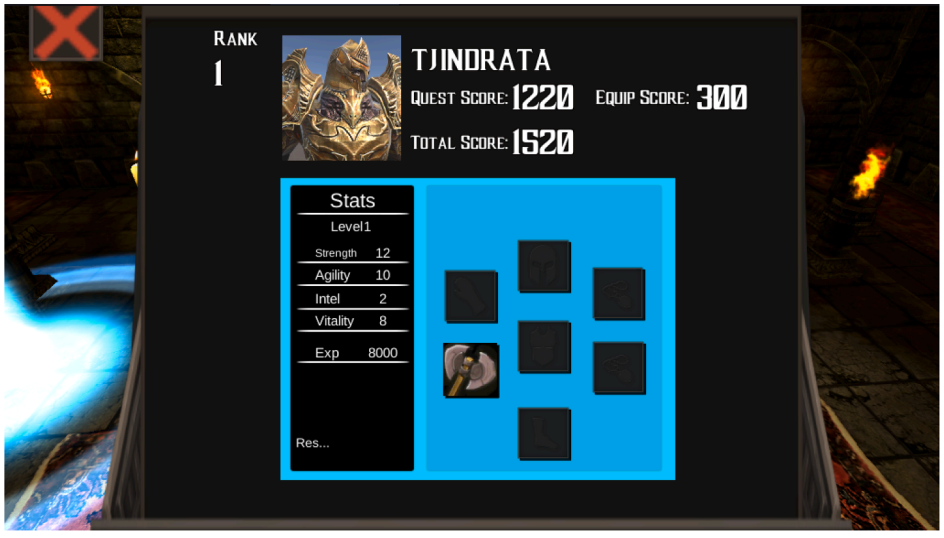
Appendix J. Pretest Results

Content	Frequency ($N = 25$)		Percentage		Conclusion
	NF	F	NF	F	
300 more points in your score. Gems can also be obtained by finishing the quest (5 gems/quest).	23	2	**92 %**	8 %	NF
300 more points in your score. Gems can also be obtained by finishing the quest (10 gems/quest).	22	3	88 %	12 %	NF
300 more points in your score. Gems can also be obtained by finishing the quest (15 gems/quest).	16	9	64 %	36 %	NF
300 more points in your score. Gems can also be obtained by finishing the quest (20 gems/quest).	14	11	56 %	44 %	NF
500 more points in your score. Gems can also be obtained by finishing the quest (5 gems/quest).	22	3	88 %	12 %	NF
500 more points in your score. Gems can also be obtained by finishing the quest (10 gems/quest).	21	4	84 %	16 %	NF
500 more points in your score. Gems can also be obtained by finishing the quest (15 gems/quest).	18	7	72 %	28 %	NF
500 more points in your score. Gems can also be obtained by finishing the quest (20 gems/quest).	15	10	60 %	40 %	NF
300 more points in your score. Gems can also be obtained by finishing the quest (40 gems/quest).	7	18	28 %	**72 %**	F
300 more points in your score. Gems can also be obtained by finishing the quest (45 gems/quest).	10	15	40 %	60 %	F
300 more points in your score. Gems can also be obtained by finishing the quest (50 gems/quest).	10	15	40 %	60 %	F
300 more points in your score. Gems can also be obtained by finishing the quest (55 gems/quest).	11	14	44 %	56 %	F
500 more points in your score. Gems can also be obtained by finishing the quest (40 gems/quest).	9	16	36 %	64 %	F
500 more points in your score. Gems can also be obtained by finishing the quest (45 gems/quest).	9	16	36 %	64 %	F
500 more points in your score. Gems can also be obtained by finishing the quest (50 gems/quest).	12	13	48 %	52 %	F
500 more points in your score. Gems can also be obtained by finishing the quest (55 gems/quest).	12	13	48 %	52 %	F

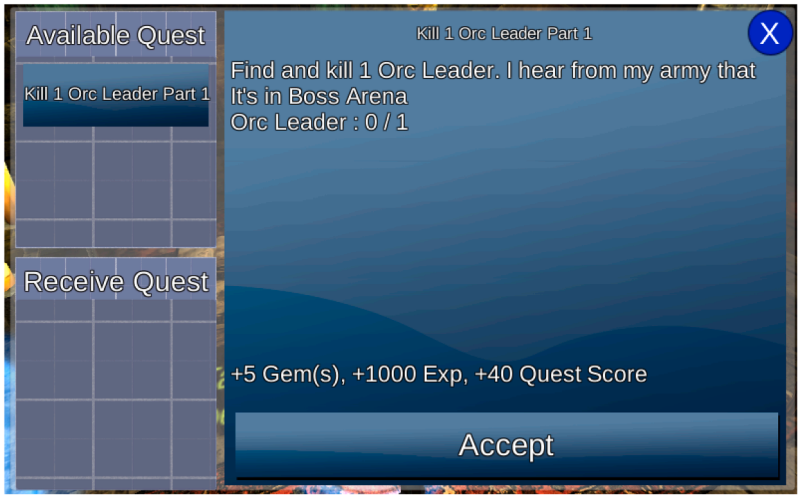
Appendix K. Gaia World (Fair Interface)

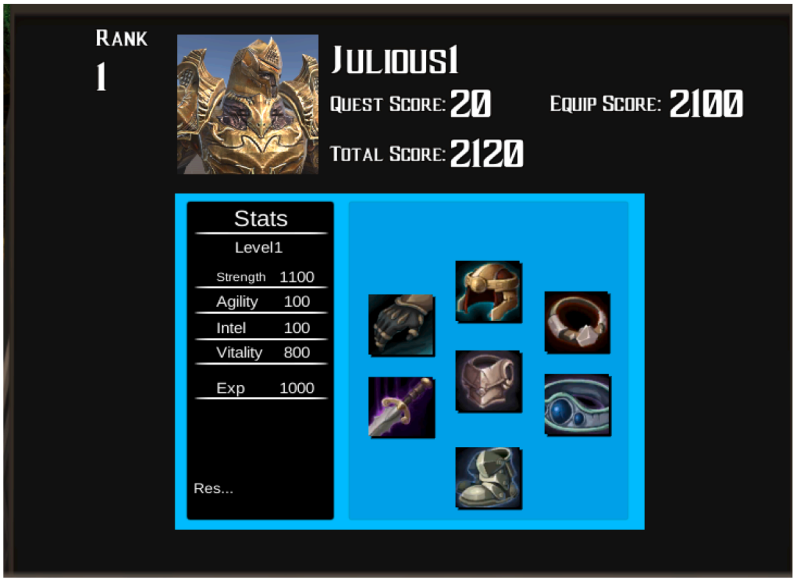
Available Quest
Kill 1 Orc Leader Part 1
Receive Quest
Kill 1 Orc Leader Part 1
Find and kill 1 Orc Leader. I hear from my army that It's in Boss Arena
Orc Leader : 0 / 1
+40 Gem(s), +1000 Exp, +40 Quest Score
Accept

1
TJINDRATA
Quest Score: 1220
Equip Score: 300
Total Score: 1520
Detail
2
JULIOUS
Quest Score: 480
Equip Score: 300
Total Score: 780
Detail
3
F1724
Quest Score: 700
Equip Score: 0
Total Score: 700
Detail
4
TANTOS
Detail


Rank
1
TJINDRATA
Quest Score: 1220
Equip Score: 300
Total Score: 1520
Stats
Level1
Strength 12
Agility 10
Intel 2
Vitality 8
Exp 8000
Res...

Appendix L. Gaia World (Not Fair Interface)





Appendix M. Results of Factor Analysis

Research Construct	Variables	Research Items	Factor Loading (>0.6)
Skill–Challenge Balance	SC1	I was challenged, but I believed my skills would allow me to meet the challenge in Gaia World.	0.756
	SC2	My abilities matched the high challenge of the game's situation in Gaia World.	0.829
	SC3	I felt I was competent enough to meet the high demands of Gaia World.	0.879
	SC4	The challenge and my skills were at an equally high level in Gaia World.	0.799
Control	CL1	I had a lot of control in Gaia World.	0.866
	CL2	I could choose freely what I wanted to see or do in Gaia World.	0.881
	CL3	I had little control over what I could do in Gaia World.*	Deleted
	CL4	I had no control over my interaction in Gaia World.*	Deleted
	CL5	I was allowed to control my interaction in Gaia World.	0.779
Social Interaction	SI1	Playing Gaia World enables me to make friends.	0.771
	SI2	I enjoy meeting the friends I made while playing Gaia World.	0.873
	SI3	Communicating with others is useful for playing Gaia World.	0.705
	SI4	Cooperating with others makes Gaia World more enjoyable.	Deleted
Perceived Enjoyment	PE1	I found playing Gaia World to be enjoyable.	0.936
	PE2	I had fun playing Gaia World.	0.928
	PE3	Playing Gaia World was boring.*	Deleted
	PE4	Gaia World annoyed me.*	Deleted
	PE5	Gaia World experience was pleasurable.	0.771
	PE6	Gaia World experience left me unsatisfied.*	Deleted
Fairness	PF1	I think Gaia World has fair policies and practices in providing in-app purchase.	0.868
	PF2	Gaia World tries to providing in-app purchase items equally.	0.862
	PF3	I feel the resource allocation is fair in Gaia World, such as virtual treasure/item, virtual money and trading modes.	0.863
	PF4	While playing Gaia World, the score I got equal with the efforts I put into.	0.712
	PF5	It felt unfair that other players have in-app purchase items and I was not.*	Deleted
Benign Envy	BE1	I felt benign envy toward the players who has in-app purchase items.	0.724
	BE2	I wanted to try harder to obtain in-app purchase items as well.	0.844
	BE3	I felt inspired to also obtain in-app purchase items.	0.877
	BE4	I wished to have in-app purchase items too.	0.805
	BE5	I wanted to be like the players who has in-app purchase items.	0.816
	BE6	I desired in-app purchase items.	0.849
Malicious Envy	ME1	I felt malicious envy toward the players who has in-app purchase items.	0.736
	ME2	I wished that other players would no longer have in-app purchase items.	0.829
	ME3	I would have liked to damage in-app purchase items.	0.814
	ME4	I wished that other players who has in-app purchase items would fail at something.	0.842
	ME5	I would have liked to hurt other players who has in-app purchase items.	0.817
	ME6	I had negative thoughts about other players who has in-app purchase items.	0.793
	ME7	I would have liked to take in-app purchase items away from other players.	0.829
In-App Purchase Intention	IAPI1	In the near future, I will consider paying for Gaia World.	0.795
	IAPI2	In the near future, the chance of my paying for Gaia World/buy in-game item is high.	0.934
	IAPI3	In the near future, I am determined to pay Gaia World/in-game item.	0.940

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